Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences

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[1] Rives A, Meier J, Sercu T, et al. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences[J]. Proceedings of the national academy of sciences, 2021, 118(15): e2016239118.

Motivation and Background

- A protein is a sequence of many residues, each chosen from 20 standard amino acids
- Goal:Predict protein structure and function from sequence
- Challenge: Long-range dependencies (far in sequence, close in 3D space)

Traditional Method

- **MSA (Multiple Sequence Alignment)**
  - Align similar sequences
  - Extract evolutionary statistics
- **HMM**
  - Probabilistic model trained from MSA
  - Captures sequence distribution within a protein family
- **Limitations**
  - Requires many related sequences
  - Mostly local modeling

Transformer method

- **Model: Transformer for Protein Sequences**
  - **Input:** amino acid sequence
  - **Architecture:** Transformer (self-attention)
  - **Output:** hidden representations $h_i \in \mathbb{R}^d$
- **Training Objective: Masked Language Modeling (MLM)**
  - Randomly mask a subset of amino acids
  - Predict missing tokens from context

$$\mathcal{L}_{\text{MLM}} = \mathbb{E}_x \, \mathbb{E}_M \sum_{i \in M} -\log p(x_i \mid x_M)$$

Power-law behavior

Table 1. Evaluation of language models for generalization to held-out UniRef50 clusters

	Model		Params	Training	ECE
(a)	Oracle				1
	Uniform Random				25
(b)	*n*-gram	4-gram		UR50/S	17.18
(c)	LSTM	Small	28.4 M	UR50/S	14.42
	LSTM	Large	113.4 M	UR50/S	13.54
(d)	Transformer	6-layer	42.6 M	UR50/S	11.79
	Transformer	12-layer	85.1 M	UR50/S	10.45
(e)	Transformer	34-layer	669.2 M	UR100	10.32
	Transformer	34-layer	669.2 M	UR50/S	8.54
	Transformer	34-layer	669.2 M	UR50/D	8.46
(f)	Transformer	10% data	669.2 M	UR50/S	10.99
	Transformer	1% data	669.2 M	UR50/S	15.01
	Transformer	0.1% data	669.2 M	UR50/S	17.50

(a) ECE ranges from 25 for a random model to 1 for a perfect model. (b) Best *n*-gram model across range of context sizes and Laplace smoothing settings. (c) State-of-the-art LSTM bidirectional language models (9). (d) Transformer model baselines with 6 and 12 layers. Small Transformer models have better performance than LSTMs despite having fewer parameters. (e) Transformers that are 34-layer models are trained on datasets of differing sequence diversity. Increasing the diversity of the training set improves generalization. High-capacity Transformer models outperform LSTMs and smaller Transformers. (f) Transformers that are 34-layer models are trained on reduced fractions of data. Increasing training data improves generalization.

- Transformer outperforms LSTM at similar scale
- Larger models (up to 650M) achieve best performance
- More diverse datasets (UR50 vs UR100) improve generalization

Biochemical Property

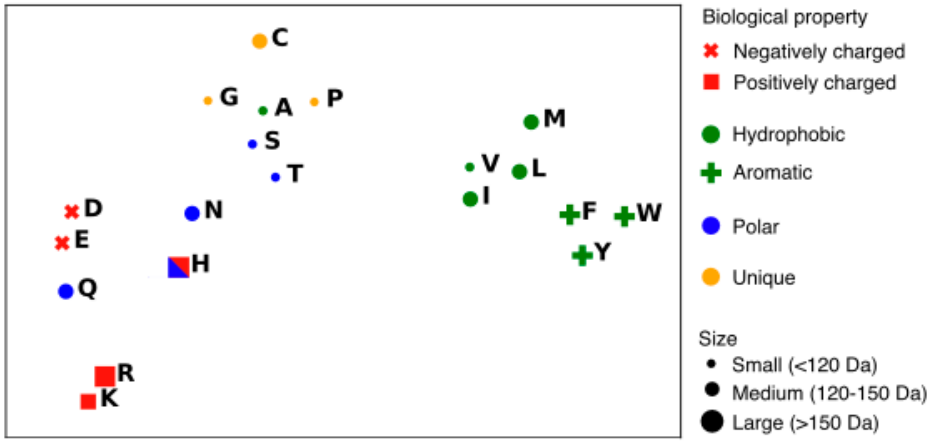

Fig. 1. Biochemical properties of amino acids are represented in the Transformer model's output embeddings, visualized here with t-SNE. Through unsupervised learning, residues are clustered into hydrophobic, polar, and aromatic groups and reflect overall organization by molecular weight and charge. Visualization of 36-layer Transformer trained on UniParc.

- Project the rows of W_{out} of the final embedding layer into two dimensions.
- Learned embeddings exhibit clustering by:
  - hydrophobicity
  - charge
  - aromaticity

Protein-level Biological Variations

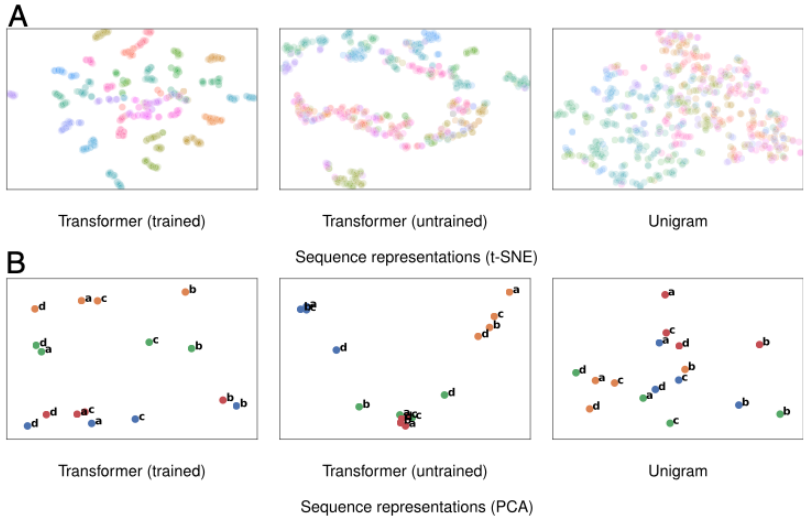

Fig. 2. Protein sequence representations encode and organize biological variations. (*A*) Each point represents a gene, and each gene is colored by the orthologous group it belongs to (dimensionality is reduced by t-SNE). Orthologous groups of genes are densely clustered in the trained representation space. By contrast, the untrained representation space and unigram representations do not reflect strong organization by evolutionary relationships. (*B*) Genes corresponding to a common biological variation are related linearly in the trained representation space. Genes are colored by their orthologous group, and their species are indicated by a character label. PCA recovers a species axis (horizontal) and orthology axis (vertical) in the trained representation space but not in the untrained or unigram spaces. Representations are from the 36-layer Transformer model trained on UniParc.

- **Method:** Represent each protein by averaging token embeddings
- **Background**: gene $\rightarrow$ (translation) $\rightarrow$ protein sequence; similar genes $\rightarrow$ similar proteins
- **Observation**
  - Orthologous proteins form clusters in representation space
  - PCA reveals meaningful biological axes (e.g., species, function)

Protein-level Remote homology detection

Table 2. Remote homology detection

		Hit-10		AUC	
	Pretraining	Fold	SF	Fold	SF
HHblits*		0.584	0.965	0.831	0.951
LSTM (S)	UR50/S	0.558	0.760	0.801	0.863
LSTM (L)	UR50/S	0.574	0.813	0.805	0.880
Transformer-6	UR50/S	0.653	0.878	0.768	0.901
Transformer-12	UR50/S	0.639	0.915	0.778	0.942
Transformer-34	(None)	0.481	0.527	0.755	0.807
Transformer-34	UR100	0.599	0.841	0.753	0.876
Transformer-34	UR50/D	0.617	0.932	0.822	0.932
Transformer-34	UR50/S	0.639	0.931	0.825	0.933
ESM-1b	UR50/S	0.532	0.913	0.770	0.880

Structural homology at the fold and superfamily (SF) level is encoded in the metric structure of the representation space. Results for unsupervised classifier based on distance between vector sequence embeddings. Hit-10 reports the probability that a remote homolog is included in the 10 nearest neighbors of the query sequence. AUC is reported for classification by distance from the query in representation space. Transformer models have higher performance than LSTMs and similar performance to HMMs at the fold level.
*HHblits (15) is a state-of-the-art HMM-based method for remote homology detection, using three iterations of sequence search.

- **Task**
  - Retrieve structurally similar proteins, even when sequence similarity is low
- **Method**
  - Compute embedding vectors $\rightarrow$ measure distance $\rightarrow$ sort $\rightarrow$ get nearest neighbors

Alignment Emerges

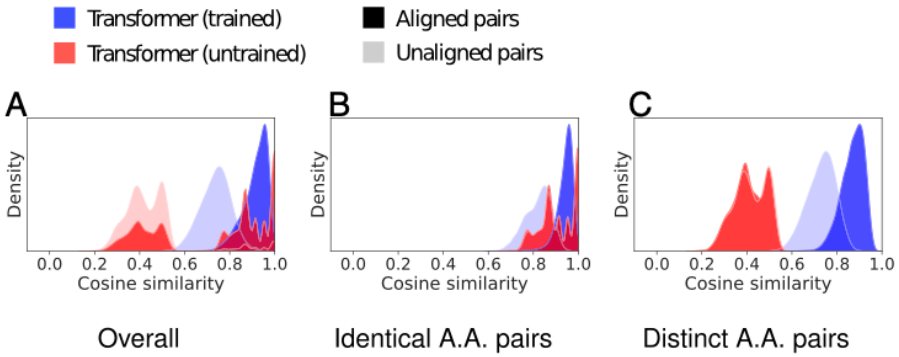

Fig. 3. Final representations from trained models implicitly align sequences. Cosine similarity distributions are depicted for the final representations of residues from sequences within Pfam family PF01010. The differences between the aligned (dark blue) and unaligned (light blue) distributions imply that the trained Transformer representations are a powerful discriminator between aligned and unaligned positions in the sequences. In contrast, representations prior to training do not separate the aligned (dark red) and unaligned positions (light red). (*A*) Overall distribution; distribution under constraint that residue pairs have (*B*) same amino acid identity; or (*C*) different amino acid identities. AUCs across 128 Pfam families are reported in *SI Appendix*, Table S1.

- **Question**
  - Do representations encode alignment (matching positions across sequences that play the same role) information?
- **Method**
  - Use ground-truth MSA (Pfam)
  - Compare cosine similarity
- **Observation**
  - Aligned pairs $\rightarrow$ higher similarity
  - Clear separation in similarity distributions
  - Holds even when amino acids differ

Linear Proj. for structure prediction

- **Background**
  - **Secondary structure:** single residue-level local patterns (e.g., α-helix, β-sheet)
  - **Tertiary structure:** 3D folding, including long-range residue contacts
  - Structure is not observed during training (latent variable)
- **Method (Linear Probe)**
  - Freeze pretrained Transformer representations h_i
  - Train simple linear models to extract structural information
- **Secondary Structure Prediction**
  - Input: residue embedding h_i
  - Model: logistic regression
  - Output: 8-class secondary structure label per position
- **Tertiary Structure (Contact Prediction)**
  - Input: pair of residues (h_i, h_j)
  - Linear projections: Wh_i, Wh_j
  - Compute score via dot product:

$$s_{ij} = (Wh_i)^\top (Wh_j)$$

  - Predict binary contact (whether residues are close in 3D)

Linear Proj. for structure prediction

Table 3. Linear projections

		SSP			Contact		
Model	Pretraining	Family	Superfamily	Fold	Family	Superfamily	Fold
HMM Profile	—	47.9 ± 1.1	47.8 ± 1.3	48.0 ± 1.6	14.8 ± 0.6	14.5 ± 1.4	14.6 ± 1.6
CCMpred	—	—	—	—	32.7 ± 1.0	32.5 ± 2.3	32.6 ± 0.4
Transformer-6	UR50/S	62.4 ± 0.7	61.8 ± 0.7	61.8 ± 1.0	24.4 ± 2.2	19.4 ± 3.1	18.6 ± 0.5
Transformer-12	UR50/S	65.5 ± 0.7	65.0 ± 1.0	65.2 ± 1.0	32.8 ± 2.9	25.8 ± 3.7	25.2 ± 0.9
Transformer-34	(None)	41.5 ± 0.9	41.3 ± 0.8	41.3 ± 1.3	8.7 ± 0.3	8.4 ± 0.7	8.4 ± 0.3
Transformer-34	UR100	65.4 ± 0.7	64.9 ± 0.9	64.9 ± 0.9	28.3 ± 2.5	22.5 ± 3.3	22.0 ± 0.8
Transformer-34	UR50/S	69.4 ± 0.6	68.8 ± 0.9	68.9 ± 0.9	43.9 ± 2.8	36.4 ± 4.2	35.3 ± 1.7
Transformer-34	UR50/D	69.5 ± 0.6	68.9 ± 0.9	69.0 ± 0.9	43.9 ± 2.8	37.1 ± 4.6	36.3 ± 2.0
ESM-1b	UR50/S	71.1 ± 0.5	70.5 ± 0.9	70.6 ± 0.9	49.2 ± 2.5	43.5 ± 4.8	42.8 ± 2.3

Fivefold cross validation experiment for generalization at the family, superfamily, and fold level. Eight-class accuracy (secondary structure), top-L long-range precision (contacts), and mean and SD across test sets for the five partitions. Minimal information about structure is present in representations prior to training. Information about secondary and tertiary structure emerges in representations as a result of unsupervised learning on sequences with the language modeling objective. Increasing diversity of sequences improves learning of structure (higher diversity UR50 datasets improve over UR100). Learned representations enable linear projections to generalize to held-out folds, outperforming projections of the sequence profile and contacts identified by the CCMpred (51) implementation of direct coupling analysis.

Result

- Strong performance after pretraining
- Structural information is already encoded in representations

DM for structure predition

- **Method**
  - Use pretrained Transformer representations h_i
  - Train **deep supervised models** on top
- **Results**
  - **Secondary structure:**
    - Transformer: $\sim$71.6% accuracy
    - Comparable to HMM-based profiles (71.2%)
  - **Contact prediction:**
    - Transformer improves over LSTM baselines
    - Still below SOTA (RaptorX with MSA features)
- **Interpretation**
  - Representations contain strong structural information
  - MSA-based methods still superior for complex long-range interactions

MSA+Transformer feature

Use MSA-based methods as the baseline, and then ADD Transformer features on top to improve prediction.

- **direct:** use final hidden representation h_i directly
- **avg:** average h_i across sequences in the MSA
- **cov:** compute pairwise features using uncentered covariance

Table 6. Feature combination (secondary structure prediction)

Features	CB513	CASP13
RaptorX	70.6	
NetSurf	72.1	74
(a) NetSurf (reimpl.)	71.2 ± 0.1	72.3 ± 0.9
(b) +direct	72.1 ± 0.1	72.2 ± 0.5
(c) +avg	73.7 ± 0.2	75.1 ± 0.4

Eight-class accuracy. The language model improves state-of-the-art features for secondary structure prediction. Features from a reimplementation of NetSurf (53) are combined with 34-layer Transformer (UR50/S) embeddings using a two-layer bidirectional LSTM architecture. (a) Performance of NetSurf features alone. (b) *Direct* adds the Transformer representation of the input sequence. (c) *Avg* adds the average of Transformer features for each position in the MSA of the input sequence. Results exceed those for state-of-the-art methods RaptorX (70.6%) and NetSurf (72.1%) on the CB513 test set and for NetSurf (74.0%) on the CASP13 evaluation set used here. Mean and SD across five random training seeds for the downstream model are shown.

Table 7. Feature combination (contact prediction)

	Test	CASP11	CASP12	CASP13
No. domains	500	105	55	34
(a) RaptorX	59.4 ± 0.2	53.8 ± 0.3	51.1 ± 0.2	43.4 ± 0.4
(b) +direct	61.7 ± 0.4	55.0 ± 0.1	51.5 ± 0.5	43.7 ± 0.4
(c) +avg	62.9 ± 0.4	56.6 ± 0.4	52.4 ± 0.5	44.8 ± 0.8
(d) +cov	63.3 ± 0.2	56.8 ± 0.2	53.0 ± 0.3	45.2 ± 0.5

Top-L long-range contact precision. The language model improves state-of-the-art features for contact prediction. A deep ResNet with fixed architecture is trained on each feature set to predict binary contacts. (a) Performance of state-of-the-art RaptorX (54) features, including PSSM, predicted secondary structure, predicted accessibility, pairwise average product correction-corrected Potts model couplings and mutual information, and a pairwise contact potential. (b) Adds Transformer representation of the input sequence to the feature set. (c) Adds the average Transformer representation at each position of the MSA. (d) Adds the uncentered covariance over the MSA of a low-dimensional projection of the Transformer features. Features are from the 34-layer Transformer pretrained on UR50/S. Mean and SD across five random training seeds for the downstream model are shown.